



DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

June 28, 2022

Surname:

Name:

Student ID:

Row:

Column:

Time: 2 hours

Maximum Marks: 33

- The following exam is composed of **8 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided either outside these spaces or **without a motivation** will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices**, such as laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme, and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above-mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above-mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.
- **Three of the points will be given on the basis on how quick you are in solving the exam.** If you finish earlier than 45 min before the end of the exam you will get 3 points, if you finish earlier than 30 min you will get 2 points and if you finish earlier than 15 min you will get 1 point (the points cannot be accumulated).
- The box on Page 10 can only be used to complete the Exercises 7, and/or 8.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Time	Tot.
/ 7	/ 7	/ 2	/ 2	/ 2	/ 2	/ 4	/ 4	/ 3	/ 33

Exercise 1 (7 marks)

Describe the Bayesian Linear Regression method and how it compares to Ridge Regression.

Exercise 2 (7 marks)

Explain what the Kernel Trick is, what it is used for, and in which ML methods it can be used.

Exercise 3 (2 marks)

Assume you have been given a dataset with input matrix X and target vector y . Consider the following snippet of code:

```
1 pca = PCA()
2 pca.fit(X, y)
3 explained = pca.explained_variance_
4 T = pca.transform(X)
5 explained_variance = np.cumsum(explained) / sum(explained)
6 T_tilde = T[:, explained_variance < 0.95]
```

1. Describe the procedure and the purpose of the above code snippet. Is it correct?
2. Line 6 consists in a selection procedure. Explain the rationale behind this operation and suggest other viable options to perform the selection procedure.
3. Do you think this code requires some preliminary operations on X and y before being executed? In the case of a positive answer, tell which ones and why should they be performed. In the case of a negative answer, motivate adequately.

1. The code is implementing the PCA algorithm for feature extraction. It starts computing the loadings matrix, and applying them to the original data input X . Finally, it selects the principal components retaining at least 95% of the variance present in the data. The issue of this code is that the fit procedure is using also the target y , while this procedure is unsupervised and therefore it does not require the real target.
2. As mentioned above, the line selects the principal components retaining at least 95% of the variance present in the data. Other viable options are the detection of an "elbow" in the cumulated variance or retaining all the principal components explaining at least $k\%$ of the variance, with a fixed parameter $k \in (0, 1)$.
3. We need to remove the empirical mean from the input otherwise the principal components might be strongly affected by it.

Exercise 4 (2 marks)

Tell if the following statements about classification algorithms are true or false. Motivate your answers.

1. The perceptron classifier and the logistic regression classifier are both generalized linear models.
2. The logistic regression loss function is convex, therefore, it admits a closed-form solution of the optimal weights.
3. If the training set is linearly separable, the logistic regression classifier and the perceptron classifier converge to the same solution.
4. The logistic regression classifier converges to a solution even when the training set is not linearly separable.

1. TRUE. Indeed, their prediction function can be both written as $f(x^T w + w_0)$. In particular, for the perceptron we have $f(y) = \begin{cases} +1 & \text{if } y \geq 0 \\ -1 & \text{otherwise} \end{cases}$ and for logistic regression we have $f(y) = \sigma(y)$, where $\sigma(\cdot)$ is the sigmoid function.
2. FALSE. While it is true that the loss function of logistic regression is convex, it does not admit a closed-form solution of the optimal weights. In general, convexity does not imply the existence of closed-form solution.
3. FALSE. Indeed, the perceptron converges to a separating hyperplane depending on the initialization, whereas logistic regression converges to the separating hyperplane that minimizes its (convex) loss function. If there exists only one separating hyperplane, the two algorithms converge to it.
4. TRUE. Indeed, logistic regression always provides the weights minimizing the corresponding loss function, being the objective function convex. If the training set is not linearly separable, the provided weights would not correspond to a separating hyperplane.

Exercise 5 (2 marks)

Tell if the following statements about the VC dimension are true or false. Motivate your answers.

1. If the VC dimension of a hypothesis space H is infinite, then hypothesis space H contains infinite hypotheses.
2. The VC dimension of a hypothesis space H is at least k if and only if the hypothesis space H shatters every subset of cardinality k of the instance space.
3. The VC dimension of the logistic regression classifier with features x_1 and x_2 is smaller than the VC dimension of the logistic regression classifier with features $x_1, x_2, x_1^2, x_2^2, x_1x_2$.
4. The VC dimension of a hypothesis space is affected by the size of the training set, but not by the size of the test set.

1. TRUE. If this were not the case, we would conclude that hypothesis space with finite cardinality would have infinite VC dimension, which is clearly false.
2. FALSE. The VC dimension is at least k if the hypothesis space shatters **at least one** subset of cardinality k in the instance space.
3. TRUE. Since the hypothesis space of the second logistic regression classifier is a superset of the hypothesis space of the first logistic regression classifier, the statement is true.
4. FALSE. The VC dimension is a property of the hypothesis space per se and does not depend on the number of samples used for training or testing.

Exercise 6 (2 marks)

You are the head of the data science department of the F1 team Beta Giulietta. You are faced with two different problems:

1. Find how the configuration of the car (i.e., flap incidence, tyres choice, aerodynamic profile) provides information about the lap time.
2. Determine the best strategy to enter the pit and change the tyres during the race. In this problem, the choice you are facing is whether to pit your car or not on each lap of the race.

Model these two problems using ML, specifying the data (input, output, state, action, reward) required, the problem class, and a specific method to solve them.

1. This scenario can be modeled as a regression problem, where the different parameters determining the configuration of the car are the input and the lap time is the target we want to model. In this case a solution might be to use Linear Regression or Gaussian Processes.
2. This scenario is a reinforcement learning problem, where the actions are pit/no-pit, the state is the consumption of the wheel, the fuel remaining in the car, the weather forecast, other cars' strategy. The reward is proportional to the position of the car at the end of the race. One might use Q-learning or SARSA to solve the problem.

Exercise 7 (4 marks)

Consider a K -NN model for regression, and the following training dataset that includes the one-dimensional feature x , the target t , and the real function $f(x)$ (unknown to the model).

x	1	1.5	3	4.5	5.5	6.5
t	2.5	2.5	6	10	10.5	12
$f(x)$	2	3	6	9	11	13

1. Compute the predictions for each element in the given dataset obtained by the model with $K = 3$.
2. Compute the mean squared error on the given dataset.
3. Provide an estimate of the variance of the model with the available data. What happens to the variance if we increase K ?
4. Provide an estimate of the squared bias of the model with the available data. What happens to the bias if we increase K ?

1. According to the K -NN model, we compute the predictions as $y(x) = \frac{1}{K} \sum_{i=1}^K t_i$, where t_i are the targets of the K nearest neighbours.

x	1	1.5	3
$y(x)$	$\frac{2.5+2.5+6}{3} = \frac{11}{3}$	$\frac{2.5+2.5+6}{3} = \frac{11}{3}$	$\frac{2.5+6+10}{3} = \frac{37}{6}$
x	4.5	5.5	6.5
$y(x)$	$\frac{6+10+10.5}{3} = \frac{53}{6}$	$\frac{10+10.5+12}{3} = \frac{65}{6}$	$\frac{10+10.5+12}{3} = \frac{65}{6}$

2. The mean squared error on the training set is given by

$$\frac{1}{N} \sum_{i=1}^N (t_i - y(x_i))^2 = \frac{(\frac{11}{3} - 2.5)^2 + (\frac{11}{3} - 2.5)^2 + (\frac{37}{6} - 6)^2 + (\frac{53}{6} - 10)^2 + (\frac{65}{6} - 10.5)^2 + (\frac{65}{6} - 12)^2}{6}$$

3. An estimate of the variance of the model is given by

$$\begin{aligned} \frac{\widehat{Var}[t]}{K} &= \frac{\frac{1}{N} \sum_{i=1}^N (t_i - f(x_i))^2}{K} \\ &= \frac{(2.5 - 2)^2 + (2.5 - 3)^2 + (6 - 6)^2 + (10 - 9)^2 + (10.5 - 11)^2 + (12 - 13)^2}{6K}, \end{aligned}$$

thus, increasing K would make the variance smaller.

4. An estimate of the bias of the model is given by

$$\frac{1}{N} \sum_{i=1}^N (f(x_i) - y(x_i))^2 = \frac{(\frac{11}{3} - 2)^2 + (\frac{11}{3} - 3)^2 + (\frac{37}{6} - 6)^2 + (\frac{53}{6} - 9)^2 + (\frac{65}{6} - 11)^2 + (\frac{65}{6} - 13)^2}{6}.$$

While the parameter K does not explicitly appears in the formula, it affects the $y(x)$ values, and we can expect the bias to increase if we increase K .

Exercise 8 (4 marks)

Consider the trajectory below, which is obtained while running the SARSA algorithm in an MDP with three states $\mathcal{S} = \{A, B, C\}$, two actions $\mathcal{A} = \{u, d\}$, and discount factor $\gamma = 1$.

$$(A, u, 2) \rightarrow (B, d, -2) \rightarrow (A, d, -2) \rightarrow (A, u, -1) \rightarrow (B, u, -3) \rightarrow (C, d, 4) \rightarrow (B)$$

1. Provide a consistent guess over the policy π that has been used to draw the latter trajectory. Do you think it is a reasonable policy?
2. Provide the policy evaluation step according to SARSA by considering zero initial value $Q(s, a) = 0$ for every state-action pair, and learning rate $\alpha = 0.5$.
3. Provide the policy improvement step, i.e., the policy π' that the algorithm will deploy at the next iteration.

1. A guess over π that is consistent with the available data is

$$\pi(u|A) = 1/3, \pi(d|A) = 2/3, \quad \pi(u|B) = 1/2, \pi(d|B) = 1/2, \quad \pi(u|C) = 0, \pi(d|C) = 1.$$

We do not have enough data to reliably estimate the policy π . On the one hand, it is unreasonable to have a deterministic policy in the state C , as we should guarantee sufficient exploration for the SARSA algorithm to converge to the optimal policy. On the other hand, the policy in the states A, B seems to be overly randomized to be an ϵ -greedy policy.

2. For the policy evaluation step, we employ the TD update formula

$$Q^\pi(s, a) \leftarrow (1 - \alpha)Q^\pi(s, a) + \alpha(R(s, a) + \gamma Q^\pi(s', a')).$$

More specifically, starting from null values for the state/action value function we have:

$$\begin{aligned} Q^\pi(A, u) &= 0.5 \cdot Q^\pi(A, u) + 0.5(2 + Q^\pi(B, d)) = 0.5 \cdot 0 + 0.5(2 + 0) = 1 \\ Q^\pi(B, d) &= 0.5 \cdot Q^\pi(B, d) + 0.5(-2 + Q^\pi(A, d)) = 0.5 \cdot 0 + 0.5(-2 + 0) = -1 \\ Q^\pi(A, d) &= 0.5 \cdot Q^\pi(A, d) + 0.5(-2 + Q^\pi(A, u)) = 0.5 \cdot 0 + 0.5(-2 + 1) = -0.5 \\ Q^\pi(A, u) &= 0.5 \cdot Q^\pi(A, u) + 0.5(2 + Q^\pi(B, u)) = 0.5 \cdot 1 + 0.5(-1 + 0) = 0 \\ Q^\pi(B, u) &= 0.5 \cdot Q^\pi(B, u) + 0.5(-3 + Q^\pi(C, d)) = 0.5 \cdot 0 + 0.5(-3 + 0) = -1.5 \\ Q^\pi(C, d) &= 0.5 \cdot Q^\pi(C, d) + 0.5(2 + 0) = 0.5 \cdot 0 + 0.5(2 + 0) = 1 \end{aligned}$$

3. With the SARSA algorithm we have to ensure sufficient exploration, e.g., by deploying an ϵ -greedy policy. Thus, we have:

$$\pi(u|A) = 1 - \epsilon, \pi(d|A) = \epsilon, \quad \pi(u|B) = \epsilon, \pi(d|B) = 1 - \epsilon, \quad \pi(u|C) = \epsilon, \pi(d|C) = 1 - \epsilon.$$

